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Started at	Sunday, April 24, 2022, 1:30 p.m
status	Completed
Finished at	Sunday, April 24, 2022, 1:36 pm
Spent time	6 minutes
feedback	Congratulations, excellent performance (90-100%).

question **1**

Complete

Achievable
points: 1.00

Name the different dimensions in which we can separate the access to allow for multiple communications in parallel?

Choose one or more answers:

- ☐ a. Orthogonal Division Multiple Access
- ☐ b. Resource Division Multiple Access
- ☒ c. Frequency Division Multiple Access
- ☒ i.e. Space Division Multiple Access
- ☒ e. Time Division Multiple Access

question **2**

Complete

Achievable
points: 1.00

Three wireless terminals are given: A, B and C. A can communicate with B only. C can communicate with B only. A sends to B, C wants to send to B. Which terminal is “hidden” in this case? At which terminal the collision may happen in this case?

Choose one or more answers:

- ☐ a. B is “hidden” for C
- ☒ b. A is “hidden” for C
- ☐ c. Collision happens at A
- ☒ i.e. C is “hidden” for A
- ☒ e. Collision happens at B

question **3**

Complete

Achievable
points: 1.00

Three wireless terminals are given: A, B and C. A can communicate with B only. C can communicate with B only. A is hidden from C. B sends to A. Which terminal is considered “exposed” to this specific communication?

Choose one or more answers:

- ☐ a. C is “exposed” to A
- ☐ b. B is “exposed” to C
- ☒ c. C is “exposed” to B

Frage **4**

Vollständig

Erreichbare
Punkte: 1,00

What are the causes of the extra problems in wireless MAC?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Changes of the signal strength due to mobility
- ☒ b. Huge variation of the signal strength over distance
- ☐ c. Larger distance between entities
- ☒ d. Much higher signal loss than wired



Frage 5

Vollständig

Erreichbare
Punkte: 1,00

What are the new problems with respect to MAC in wireless systems?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Multiple devices compete for a common, shared resource
- ☐ b. The physical layer may produce transmission errors
- ☒ c. The sender and receiver do not have a consistent view on the radio resource
- ☒ d. Collisions happening at the receiver may not be noted by the sender

Frage 6

Vollständig

Erreichbare
Punkte: 1,00

What does Media Access Control deal with?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Control the transmit power of the sender
- ☐ b. Static multiplexing of frequency, time and spatial resources
- ☒ c. Avoid conflicts between multiple senders
- ☒ d. Schedule a common radio resource dynamically
- ☒ e. Control the dynamic access of senders to the medium

Frage 7

Vollständig

Erreichbare
Punkte: 1,00

What is the cause of the near-far terminal problem?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The nearest receiver is “drowned out” by the signal of the transmitter
- ☒ b. Due to the decrease of signal strength with distance the receiver will receive more power from the nearest transmitter
- ☐ c. Due to the decrease of signal strength with distance the transmitter will need more power to reach the nearest receiver
- ☒ d. The nearest transmitter “drowns out” the signal from the farthest transmitter

Frage 8

Vollständig

Erreichbare
Punkte: 1,00

What is the responsibility of the Media Access Control sublayer in communication systems?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Error correction
- ☒ b. Addressing
- ☐ c. Modulation
- ☒ d. Control for accessing the transmission medium
- ☒ e. Arbitration of access from different devices

Frage 9

Vollständig

Erreichbare
Punkte: 1,00

What problems may happen with sensing and collision detection in wireless networks?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Collision detection may not report a collision though it happens
- ☒ b. Sensing may report a busy channel though it is free
- ☒ c. Collision detection may report a collision though none happens
- ☒ d. Sensing may report a free channel though it is busy

Frage 10

Vollständig

Erreichbare
Punkte: 1,00

Why does CSMA/CD known from the Ethernet fails in wireless networks?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Carrier Sensing may not work because of the Hidden Terminal problem
- ☒ b. Because Collision Detection would be needed at the receiver and not at the sender side
- ☒ c. Carrier Sensing (CS) and Collision Detection (CD) may not work because of high signal attenuation in wireless networks
- ☒ d. Because Carrier Sensing would be needed at the receiver and not at the sender

Aktivität

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